

ROSE WABERE

Data and Analytics Engineer | Python SQL PySpark dbt Airflow Power BI | ETL Pipelines ML Systems RAG Governance Analytics

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PROFESSIONAL SUMMARY

Data and Analytics Engineer with 4+ years of hands-on experience designing and delivering production-grade ETL/ELT pipelines, analytics-ready datasets, and ML-integrated data systems. Improved dataset reliability by 25%, accelerated data refresh speeds by 40%, and delivered systems processing 50GB+ monthly data across AI, e-commerce, environmental, and public sector environments. Proven across Python, SQL, PySpark, dbt, Apache Airflow, and Power BI, with additional capability in LangChain RAG systems, vector databases, and ML model deployment. Brings a rare combination of engineering depth, machine learning capability, and governance-aware data thinking shaped by a criminology foundation in risk, ethics, and behavioral analytics.

TECHNICAL SKILLS

Data Engineering and Pipelines: Python, SQL (CTEs, window functions, complex joins, query optimization), PySpark, dbt, Apache Airflow, ETL/ELT Pipeline Design, API Integration, Batch Processing, Workflow Orchestration, Docker

Analytics and Business Intelligence: Power BI (Desktop and Cloud), DAX, Power Query (M), Streamlit, Grafana, KPI Dashboard Development, Dimensional Modeling, Star/Snowflake Schema Design

Data Quality and Governance: Automated Data Quality Testing and Validation, Data Lineage and Documentation, Compliance-Aware Data Handling, Audit-Ready Reporting, Records Governance

Machine Learning and AI: XGBoost, Scikit-learn, Regression and Classification, Time-Series Forecasting, Feature Engineering, LangChain, RAG Systems, Weaviate, LLM API Integration, OpenAI Function Calling

Databases, Cloud and DevOps: PostgreSQL, MySQL, MongoDB, SQLite | GCP, Vertex AI, AWS, Azure (working) | Git, FastAPI, REST APIs, Vercel, Docker

Microsoft Ecosystem: Power BI, Power Apps, Power Automate, Microsoft 365, Excel

PROFESSIONAL EXPERIENCE

Data Science and Data Engineering Associate

Data Science East Africa | Nairobi, Kenya | Jan 2025 - Present | Full-Time

- Designed and maintained Python and SQL-based ETL analytics pipelines processing 50GB+ monthly data, supporting production AI and business intelligence projects across multiple client engagements.
- Built automated data validation and quality review workflows across 15+ datasets, improving dataset reliability by 25% for downstream ML model training and forecasting.
- Developed Power BI and Streamlit dashboards monitoring KPIs, financial metrics, and operational performance, enabling real-time decision-making for management and clients.
- Translated complex business requirements from technical and non-technical stakeholders into structured, analytics-ready data models supporting seamless dashboarding and self-service reporting.
- Standardized ETL and reporting procedures; maintained comprehensive documentation for datasets, transformations, and data lineage ensuring team knowledge continuity.
- Mentored junior team members on data pipeline best practices, data quality standards, and structured transformation workflows.

Freelance AI Trainer, Data Annotator and Analyst

Remote - Global AI Clients | 2021 - Present | Part-Time

- Annotated and quality-controlled 15,000+ labeled records for production ML models, maintaining 99% accuracy through strict governance and validation protocols.
- Designed pipeline-ready datasets and executed deep-dive analyses using SQL, Python, and Excel for AI model evaluation and enterprise analytics use cases.

- Delivered structured analytical reports summarizing data quality metrics, validation outcomes, and performance diagnostics to guide client decision-making.

Sales Team Lead

King's Electronics | Nairobi, Kenya | Aug 2021 - Mar 2022 | Full-Time

- Increased monthly sales by 15% through data-driven customer targeting strategies informed by structured customer feedback analysis.
- Built Excel-based KPI tracking dashboards enabling data-backed weekly management reporting.

Case File Analyst (Intern)

Kamukunji Police Station | Nairobi, Kenya | Oct 2024 - Jan 2025

- Conducted structured analysis of 200+ investigative case files, identifying behavioral and crime pattern trends supporting operational planning decisions.
- Digitized and catalogued sensitive records while maintaining strict data confidentiality, chain-of-custody, and governance standards.

Inventory and Customer Research Analyst

Jumia Kenya / Jamboshop / Jiji | Nairobi, Kenya | Jan 2023 - 2025 | Part-Time

- Analyzed customer demand and sales data across 3 e-commerce platforms; generated reports increasing restocking accuracy by 20% and reducing peak-season stockouts.
- Built SQL and Excel reporting tools tracking metrics across 50+ product categories, delivering data-backed pricing and inventory recommendations.

Additional Roles

Emergency Response Data Intern (NDOC, Mar-Jun 2021) | Sales and Admin Assistant (Njowa Gen Traders, Mar 2020-Feb 2021) | Credit Analyst Intern (Taifa Sacco, May-Jul 2019)

- NDOC: Collected, validated, and analyzed incident data for high-stakes operational and coordination reporting in emergency environments.
- Njowa Gen Traders: Financial records, invoicing, inventory tracking, and customer documentation management.
- Taifa Sacco: Loan portfolio analysis and credit risk assessments supporting compliance reviews for 50+ clients weekly.

PROJECTS

Kipaji Chetu- Adaptive Learning Platform with AI Feedback

Lead Backend Engineer and AI Systems Developer | FastAPI, PostgreSQL, Groq AI, Edge-TTS, Whisper STT, Bootstrap, Chart.js

- Engineered a production-ready async FastAPI backend with a 7-table PostgreSQL database (students, topics, quizzes, attempts, performance logs, AI feedback logs, teacher notes) supporting a full adaptive learning ecosystem.
- Built AI-powered adaptive quiz engine using Groq LLM: dynamically generates questions calibrated to each student's difficulty level, records attempts, and delivers instant personalized AI feedback per answer submission.
- Implemented full accessibility stack: Edge-TTS audio synthesis and Groq Whisper speech-to-text transcription enabling voice-first learning for students with accessibility needs.
- Developed teacher analytics dashboard with class-wide performance stats, struggling student identification, topic difficulty heatmaps, real-time charts, and bonus STT/TTS diagnostic tools.

All API endpoints fully tested: `/api/submit-answer`, `/api/next-question` (adaptive), `/api/student-progress`, `/api/teacher-report`, `/api/tts`, `/api/stt`

Safari Scouts – AI Kenyan Travel Assistant

AI Engineer & Full-Stack Developer | Python, LangChain, Weaviate, FastAPI, Vercel

- Architected a full RAG data pipeline integrating web scraping, PDF parsing, and Weaviate vector database indexing, processing 500+ documents into a semantically searchable analytics layer.
- Designed ETL workflows automating document ingestion and transformation, reducing manual data processing by 80%; integrated LangChain LLM workflows with FastAPI, achieving sub-3-second response times.

github.com/Rozieroz/Travel-Assistant | Live: safari-scouts.vercel.app

Nairobi House Price Prediction – End-to-End ML Pipeline

Data Scientist & Full-Stack Developer | Python, XGBoost, Pandas, Streamlit, Scikit-learn

- Built complete analytics engineering pipeline: scraped 800+ property listings, cleaned and transformed data, engineered features, trained XGBoost model achieving MAE of 15.2M KSh.
- Deployed analytics outputs into interactive Streamlit dashboard enabling real-time self-service property price exploration. github.com/Rozieroz/Nairobi-House-Price-Prediction | Live: nairobi-house-price-prediction.streamlit.app

Kijani Care 360 – Community Tree Planting & Monitoring Platform

Lead Data Engineer & Full-Stack Developer | React/Vite, FastAPI, LangChain/Groq, PostgreSQL, Redis, Docker

- Engineered backend data models and analytics modules integrating environmental and geospatial datasets to score planting suitability by region.
- Automated ETL pipelines ensuring clean, query-ready datasets powering ongoing analytical dashboards.

github.com/Rozieroz/kijani_care_360 | Live: kijanicare-360.vercel.app

Africa Energy Data 2000–2024

Data Engineer & Senior Data Analyst | Python, Pandas, Streamlit, SQL

- Processed continent-wide energy datasets spanning 54 African countries and 24 years, uncovering renewable energy trends that supported 2 academic publications.
- Automated full ETL pipeline using Python and Pandas, improving data refresh speed by 40%.

github.com/Rozieroz/Africa_Energy_Data-2000-2024-

Earthquake Data Pipeline (ETL & Real-Time Visualization)

Data Engineer & Automation Developer | Python, SQLAlchemy, PostgreSQL, Apache Airflow, Grafana

- Built Python ETL pipeline extracting real-time global earthquake data from APIs into PostgreSQL with automated Airflow orchestration and Grafana monitoring dashboard.

github.com/Rozieroz/Earthquake_Data_Pipeline

Kenyan Food Price Tracker & Inflation Analysis

Data Engineer & Analyst | PySpark, PostgreSQL, Grafana, Pandas

- Built PySpark pipeline to scrape, clean, and analyze Kenyan food price data with time-series inflation forecasting and real-time Grafana economic intelligence dashboards.

github.com/Rozieroz/Kenyan-Food-Price-Tracker-and-Inflation-Analysis

Market Research AI Assistant

AI Engineer & Data Scientist | Python, LangChain, OpenAI API, FastAPI

- Built LangChain and OpenAI Function Calling pipeline processing 1,000+ market research documents, reducing client research time by 60% and identifying 12 new market niches.

github.com/Rozieroz/Market_Research_Assistant_AI_Model

EDUCATION

B.A. Security Studies and Criminology - Mount Kenya University | Nairobi, Kenya

Diploma in Criminology and Social Order - University of Nairobi | Nairobi, Kenya

Diploma in Microfinance - The Co-operative University of Kenya | Nairobi, Kenya

CERTIFICATIONS

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- [Data Analytics, Data Science and AI](#) - LuxDevHQ
 - Software Development - Women Techsters Initiative
 - Cybersecurity Fundamentals - Women Techsters Initiative
 - Green Digital Skills - INCO Academy
 - Mediation Certification - Peacecum Mediation Centre